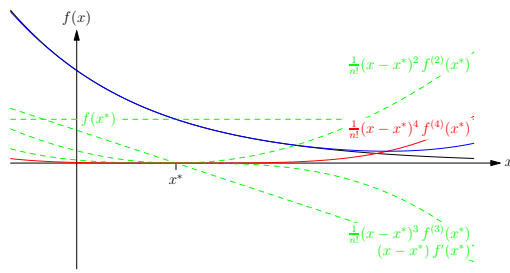


Taylor's Expansion

Converge, residue, high-dimensions, Hessian

Approximating Functions

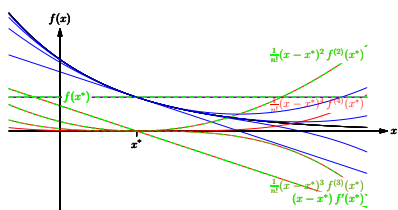
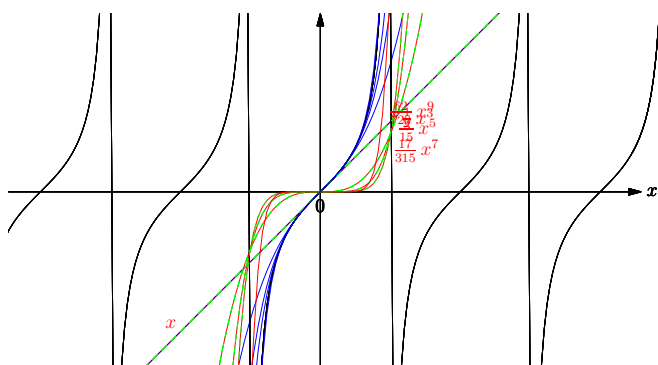
- In much of machine learning we are trying to approximate functions
- There are many different ways to do this
- Depending on your application some are much better than others
- All function approximation makes some assumptions about the regularity of the function they are approximating
- Taylor expansions are polynomial approximations of a smooth function around a point

Taylor Expansion

- This gives the Taylor expansion

$$f(x) \approx f(x^*) + f'(x^*)(x - x^*) + \frac{1}{2}f''(x^*)(x - x^*)^2 + \dots + \frac{1}{n!}f^{(n)}(x^*)(x - x^*)^n$$

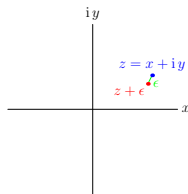
- The left and right-hand side both have the same n derivatives at the point x^*

**Taylor Expansion of $\tan(x)$** 

Analytic Functions

- An important class of functions are those where for almost all z the derivative

$$\frac{df(z)}{dz} = \lim_{\epsilon \rightarrow 0} \frac{f(z + \epsilon) - f(z)}{\epsilon}$$



does not depend on the direction ϵ

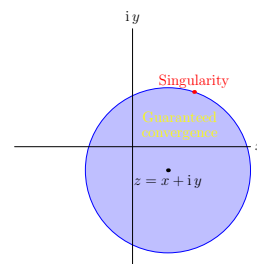
- This includes polynomials, but also includes exponentials, trigonometric functions, logarithms, ratios of polynomials, "special functions", etc.

The Hyperbolic Tangent

- The hyperbolic tangent, $\tanh$, is related to the normal tangent function through $\tanh(x) = i \tan(ix)$
- The function (along the real line) doesn't have any singularities
- But we know that $\tan(x)$ has a singularities at $\frac{\pi}{2} \pm k\pi$ where k is any integer
- Thus a Taylor expansion of $\tanh(x)$ around $x = 0$ is guaranteed to converge in a circle of radius $\pi/2$ but not necessarily for larger values of x

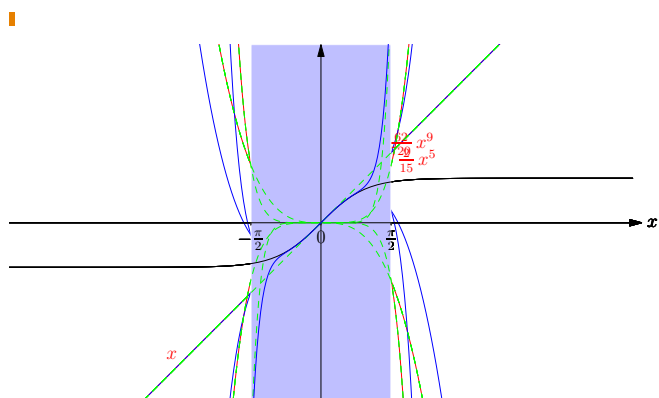
Guaranteed Convergences of Taylor Functions

- For analytic functions, Taylor series are guaranteed to converge inside a circle that excludes singular points



- E.g. the function $f(z) = \frac{1}{z^2+1}$ has a singularity at $z = \pm i$. The Taylor series of $f(z)$ around 0 is guaranteed to converge inside a circle of radius 1

Taylor Expansion of $\tanh(x)$



Residue

- Taylor expansions actually come with an error bound

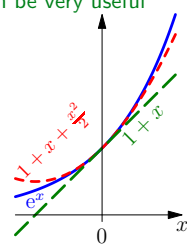
$$f(x) = \sum_{i=0}^{n-1} \frac{(x-x^*)^i}{i!} f^{(i)}(x^*) + \frac{(x-x^*)^n}{n!} f^{(n)}(\theta_x)$$

θ_x is a number in the interval between x^* and x

- The last term known as the **residue** can often be very useful

$$e^x = 1 + x + \frac{x^2}{2} e^{\theta_x}$$

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} e^{\theta_x}$$



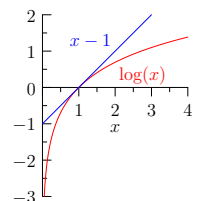
Useful Bounds

- A much used bound in mathematical analysis is $e^x \geq 1 + x$
- Many other bounds can be obtained similarly, e.g. expanding $\log(x)$ around $x = 1$

$$\log(x) = (x-1) - \frac{(x-1)^2}{2} \frac{1}{\theta^2}$$

giving

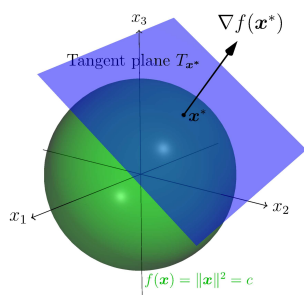
$$\log(x) \leq x-1 \quad \text{or} \quad \log(1+x) \leq x$$



- The proof of the residue for Taylors come from applying multiple integrals to the term $f^{(n)}(x)$. For the sake of time I will skip this

Outline

- 1-D Taylor Expansion
- High Dimensions
- Hessians



High Dimensional Spaces

- Just as we defined a Taylor expansion by approximating the function to a polynomial we can do the same for a function $f: \mathbb{R}^n \rightarrow \mathbb{R}$
- We can determine the coefficients by differentiating both sides and setting $x = x^*$

$$f(x^*) = c^{(0)}$$

$$\frac{\partial f(x^*)}{\partial x_i} = c_i^{(1)}$$

$$\frac{\partial^2 f(x^*)}{\partial x_i \partial x_j} = 2c_{ij}^{(2)}$$

- We find

$$f(\mathbf{x}) \approx f(\mathbf{x}^*) + (\mathbf{x} - \mathbf{x}^*)^T \nabla f(\mathbf{x}^*) + \frac{1}{2}(\mathbf{x} - \mathbf{x}^*)^T \mathbf{H}(\mathbf{x} - \mathbf{x}^*) + \dots$$

- Where the matrix $\mathbf{H}$ is known as the Hessian and has components

$$H_{ij} = \frac{\partial^2 f(\mathbf{x}^*)}{\partial x_i \partial x_j}$$

- Because

$$\frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j} = \frac{\partial^2 f(\mathbf{x})}{\partial x_j \partial x_i}$$

the Hessian is symmetric ($\mathbf{H} = \mathbf{H}^T$)

- For smooth functions in n dimensions the equation $f(\mathbf{x}) = c$ defines a $(n-1)$ -dimensional hyper-surface (because we have imposed one constraint)
- This can be thought of as the high dimensional analogue of a contour line
- The plane that goes through a point $\mathbf{x}^*$ that lines up with the contour surface is known as the tangent plane
- Away from $\mathbf{x}^*$ the value of the function $f(\mathbf{x})$ differs from the tangent plane by a quantity of order $\|\mathbf{x} - \mathbf{x}^*\|^2$

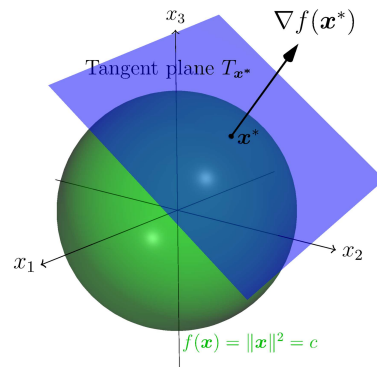
Defining the Tangent Plane

- Thus if $\mathbf{x}$ is in the tangent plane $T_{\mathbf{x}^*}$ then

$$f(\mathbf{x}) \approx f(\mathbf{x}^*) + O((\mathbf{x} - \mathbf{x}^*)^2)$$

- Or $(\mathbf{x} - \mathbf{x}^*)^T \nabla f(\mathbf{x}^*) = 0$
- That is the tangent plane at a point $\mathbf{x}^*$ is the set of points orthogonal to the gradient
- Notice that the gradient then points in the direction orthogonal to the tangent plane of the contour surface

Picture of Tangent Plane for $f(\mathbf{x}) = \|\mathbf{x}\|^2 = c$



Direction of Steepest Ascent

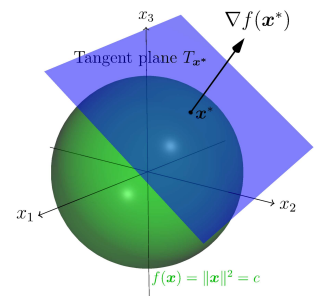
- We note that as

$$f(\mathbf{x}^* + \epsilon \mathbf{u}) = f(\mathbf{x}^*) + \epsilon \mathbf{u}^T \nabla f(\mathbf{x}^*) + O(\epsilon^2)$$

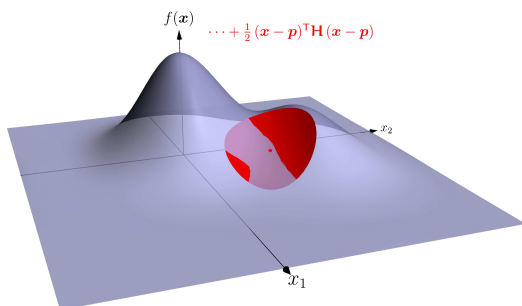
- The direction where local the function grows fastest is when $\mathbf{u}$ is in the direction $\nabla f(\mathbf{x}^*)$
- Likewise when $\mathbf{u}$ is in the direction $-\nabla f(\mathbf{x}^*)$ the function decreases as fast as possible
- This is only true locally (when $\epsilon \gg \epsilon^2$)

Outline

1. 1-D Taylor Expansion
2. High Dimensions
3. Hessians



2-D Taylor Expansion



The Hessian

- The second order term in the Taylor expansion tells us about the curvature of the function
- This is very relevant when we are at an extremal point, $\mathbf{x}^*$, where $\nabla f(\mathbf{x}^*) = 0$ so that

$$f(\mathbf{x}) = f(\mathbf{x}^*) + \frac{1}{2}(\mathbf{x} - \mathbf{x}^*)^T \mathbf{H}(\mathbf{x} - \mathbf{x}^*) + O(\|\mathbf{x} - \mathbf{x}^*\|^3)$$

- If we make a perturbation $\mathbf{x} = \mathbf{x}^* + \epsilon \mathbf{u}$ then

$$f(\mathbf{x}^* + \epsilon \mathbf{u}) = f(\mathbf{x}^*) + \frac{1}{2}\epsilon^2 \mathbf{u}^T \mathbf{H} \mathbf{u} + O(\epsilon^3)$$

- For sufficiently small perturbation the sign of change in the function depends on the sign of $\mathbf{u}^T \mathbf{H} \mathbf{u}$

Eigenvalues

- We will revisit eigen-systems, but recall that for a square matrix an eigenvector, v_i , of a matrix H is defined such that

$$Hv_i = \lambda_i v_i$$

- For an $n \times n$ symmetric matrices (such as the Hessian), the eigenvalues are all real, and we can choose n orthogonal eigenvectors such that $v_i^T v_j = \delta_{ij} \iff [i = j]$
- We note that

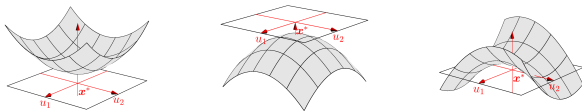
$$v_i^T H v_i = v_i^T \lambda_i v_i = \lambda_i v_i^T v_i = \lambda_i$$

- If we perturb in a direction v_i then

$$f(x^* + \epsilon v_i) = f(x^*) + \frac{1}{2} \epsilon^2 \lambda_i + O(\epsilon^3)$$

Minima, Maxima and Saddle-Points

- If all the eigenvalues are positive, $\lambda_i > 0$, then for any u we have $u^T H u > 0$ and x^* would be a local minimum. We say H is **positive definite** or $H \succ 0$
- Similarly if all the eigenvalues are less than zero (H is **negative definite**, $H \prec 0$), then x^* will be a local maximum
- Otherwise x^* is a saddle-point



Gradient Descent

- We usually proceed iteratively, starting from some $w^{(0)}$

$$w^{(t+1)} = w^{(t)} - r \nabla L(w^{(t)})$$

- This moves us in the direction that (locally) minimises the loss function $L(w)$
- As we get closer to the minimum $\nabla L(w^{(t)})$ typically becomes smaller and we take smaller steps
- However, many nasty things can happen, and we have to be careful in how we perform gradient descent
- We will revisit this when we look at optimisation

General Directions

- We can write any vector u as a weighted sum of the eigenvectors (the eigenvectors form a basis) $u = \sum_{i=1}^n c_i v_i$
- Note that

$$\begin{aligned} u^T H u &= \left(\sum_{i=1}^n c_i v_i^T \right) H \left(\sum_{j=1}^n c_j v_j \right) \\ &= \left(\sum_{i=1}^n c_i v_i^T \right) \left(\sum_{j=1}^n c_j \lambda_j v_j \right) \\ &= \sum_{i,j=1}^n \lambda_j c_i c_j v_i^T v_j = \sum_{i=1}^n \lambda_i c_i^2 \end{aligned}$$

using the orthogonality of eigenvectors $v_i^T v_j = \delta_{ij}$

Finding Optima

- We are often given a loss function, $L(w)$, which we wish to minimise
- The condition for w^* to be a minimum is that $\nabla L(w^*) = 0$ and $H \succ 0$ where H is the Hessian of $L(w)$ at the point w^*
- Except for a few very simple loss functions, there are likely to be many (and often a huge number) of local minima
- Unless $L(w)$ is very simple we are unlikely to be able to solve $\nabla L(w) = 0$ analytically

Lessons

- For functions that are continuous (as least almost everywhere) Taylor series are really useful in understanding the local behaviour of the function
- The gradient defines the direction in which a function increases fastest
- This will be orthogonal to the tangent plane, where locally the function is a constant
- The Hessian (not to be mistaken with the Jacobian) is the matrices of second derivative defining the curvature away from the tangent plane